RAJALAKSHMI ENGINEERING COLLEGE

Alagappan R 251801018-2025-AIDS A2

Started on	Thursday, 17 September 2026, 10:15 AM
State	Finished
Completed on	Thursday, 17 September 2026, 10:25 AM
Time taken	9 mins 57 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int c = 1;
    int outer = n - n / 2;

    int jcount = 0;
    int j = 1;

    while (j < n)
    {
        jcount++;
        j = j * 2;
    }

    c += outer + 1;
    c += outer * (jcount + 1);
    c += outer * jcount * (jcount + 1);
    c += outer * jcount * jcount;

    printf("%d", c);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔

	Input	Expected	Got	
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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